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stops learning
is old, whether
at twenty
or eighty.

—Henry Ford

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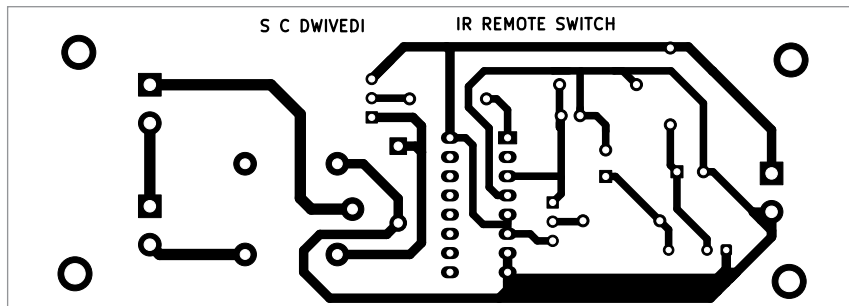


Fig. 3: Actual-size PCB layout

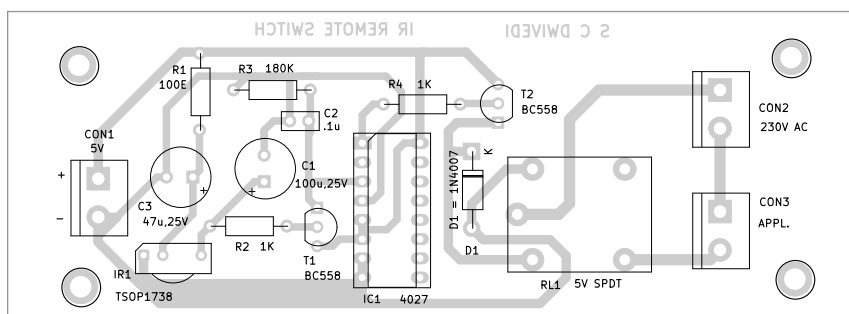


Fig. 4: Component layout

PARTS LIST

Semiconductors:

IC1 (IC1A-IC1C) - 4027 JK flip-flop
T1, T2 - BC558 PNP transistor
D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 100 Ω
R2, R4 - 1k Ω
R3 - 180k Ω

Capacitors:

C1 - 100 μ F, 25V electrolytic
C2 - 0.1 μ F ceramic disc
C3 - 47 μ F, 25V electrolytic

Miscellaneous:

CON1-CON3 - 2-pin connector
IR1 - TSOP1738 sensor module
- 16-pin IC base
- 5V battery or adaptor

Construction and testing

An actual-size, single-sided PCB layout for the IR remote switch is shown in Fig. 3, with the component layout in Fig. 4. Once assembled on the PCB, the circuit should be enclosed in a suitable cabinet with an opening for the IR module.

Alternatively, a 7cm $\times$ 9cm Vero-board can be used. All components should be soldered as per the circuit diagram using a low-wattage soldering

iron. The IC should be inserted only after soldering is complete. A 16-pin IC socket should be used for IC1, ensuring it is not placed in the socket during soldering. Lastly, the IR module should be soldered, and two-pin connectors should be attached for the 5V power supply, appliance, and 230V AC input.

After completing the assembly, connect a 5V power supply to CON1. For testing, connect a 230V, 45-watt lamp as the load and apply a 230V power supply across CON2 and CON3. Press any button on the IR remote while pointing it at the TSOP1738 IR sensor. The first press will turn the lamp on, while the second press will turn it off, repeating in a cycle.

The operating range of this switch is approximately 4 to 5 metres. Once tested successfully, the device is ready for use. **EFY**



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